EXPERIMENT NO. 05

AIM: Save user data locally using SharedPreferences (Android)

THEORY:

SharedPreferences

- Lightweight storage mechanism in Android
- Stores data as key-value pairs

Data Persistence

- Data remains saved even after app is closed
- Useful for small user data like name, settings

Saving Data

- Uses `SharedPreferences.Editor`
- Methods like `putString()`, `apply()`

Retrieving Data

- Uses methods like `getString()`
- Default value returned if key not found

Clearing Data

- `clear()` method removes all stored values

MainActivity.kt:

```
package com.example.myapplication

import android.content.Context
import android.os.Bundle
import android.widget.*
import
androidx.appcompat.app.AppCompatActi
vity

class MainActivity :
AppCompatActivity() {

   private val PREF_NAME = "MyPrefs"
   private val KEY_NAME = "name"

   override fun
onCreate(savedInstanceState:
Bundle?) {

super.onCreate(savedInstanceState)

setContentView(R.layout.activity_mai
n)

       val etName =
findViewById<EditText>(R.id.etName)
       val btnSave =
findViewById<Button>(R.id.btnSave)
       val btnLoad =
findViewById<Button>(R.id.btnLoad)
       val btnClear =
findViewById<Button>(R.id.btnClear)
       val tvResult =
findViewById<TextView>(R.id.tvResult
)

       val sharedPref =
getSharedPreferences(PREF_NAME,
Context.MODE_PRIVATE)

       // SAVE DATA
       btnSave.setOnClickListener {
           val name =
etName.text.toString()

           if (name.isEmpty()) {
               Toast.makeText(this,
"Enter name first",
Toast.LENGTH_SHORT).show()
```

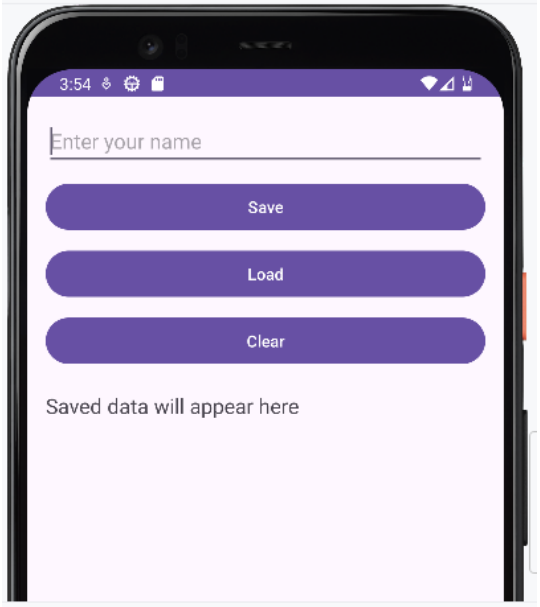
```
            } else {
                val editor =
sharedPref.edit()

editor.putString(KEY_NAME, name)
                editor.apply()

                Toast.makeText(this,
"Data Saved",
Toast.LENGTH_SHORT).show()
                etName.text.clear()
            }
        }
        // LOAD DATA
        btnLoad.setOnClickListener {
            val savedName =
sharedPref.getString(KEY_NAME, "No
data found")
            tvResult.text = "Saved
Name: $savedName"
        }

        // CLEAR DATA
        btnClear.setOnClickListener {
            val editor =
sharedPref.edit()
            editor.clear()
            editor.apply()

            tvResult.text = "Data
Cleared"
            Toast.makeText(this,
"Preferences Cleared",
Toast.LENGTH_SHORT).show()
        }
    }
}
```

activity_main.xml:

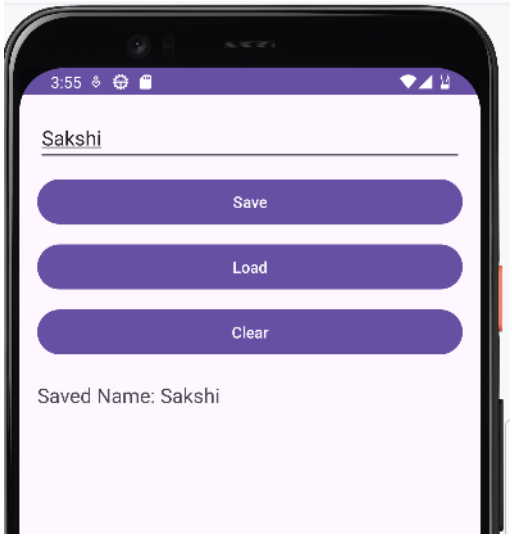
```
<?xml version="1.0"
encoding="utf-8"?>
<LinearLayout

xmlns:android="http://schemas.androi
d.com/apk/res/android"
   android:id="@+id/mainLayout"

android:layout_width="match_parent"

android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="16dp">

   <EditText
       android:id="@+id/etName"

android:layout_width="match_parent"

android:layout_height="wrap_content"
       android:hint="Enter your
name"/>

   <Button
       android:id="@+id/btnSave"

android:layout_width="match_parent"

android:layout_height="wrap_content"
       android:text="Save"

android:layout_marginTop="10dp"/>

   <Button
       android:id="@+id/btnLoad"

android:layout_width="match_parent"

android:layout_height="wrap_content"
       android:text="Load"

android:layout_marginTop="10dp"/>

   <Button
       android:id="@+id/btnClear"

android:layout_width="match_parent"

android:layout_height="wrap_content"
       android:text="Clear"

android:layout_marginTop="10dp"/>

   <TextView
       android:id="@+id/tvResult"

android:layout_width="match_parent"
```

```
android:layout_height="wrap_content"
        android:text="Saved data will
appear here"
        android:textSize="18sp"
android:layout_marginTop="20dp"/>

</LinearLayout>
```

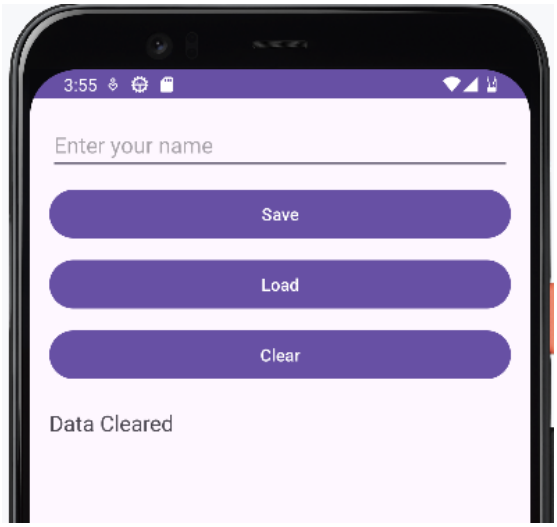
Home screen



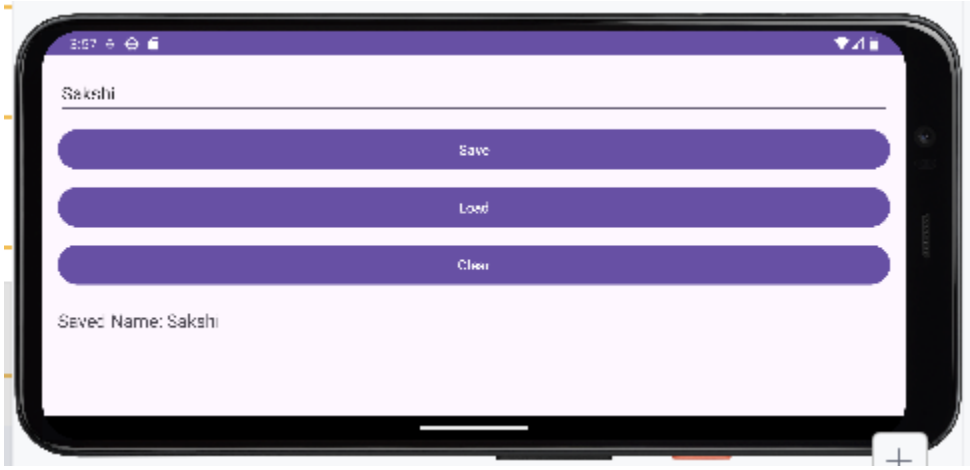
Type data and save it then load



click on clear



Saved data is loaded on to screen after rotating the phone



CONCLUSION:

This experiment demonstrates how to store, retrieve, and manage user data locally using SharedPreferences, enabling persistent storage in Android applications.